

4.8 Setting of Running Tensions

The machine winding system is designed such that the tension applied to the substrate at the unwind, coating and the rewind sections of the winding system can all be at different set-points.

The value of this web tension set-point depends on the type, thickness and width of the substrate, the physical condition of the unwind reel and the required condition of the rewind reel.

It is suggested that the starting values listed below are used initially as starting tension set-points.

Experimentation with the substrate will lead to normal operating conditions to be determined.

Typical starting values are:

Material	Newtons/ linear 1000mm of web width
12 micron Polyester	31-40
20 micron BOPP	36-44
12 micron Nylon	31-40
25 micron CPP	3.2 - 8

Table 4.8-1 Tension Guidance Levels

Once the values have been calculated, the required set-point values can be entered into the RECIPE screen.

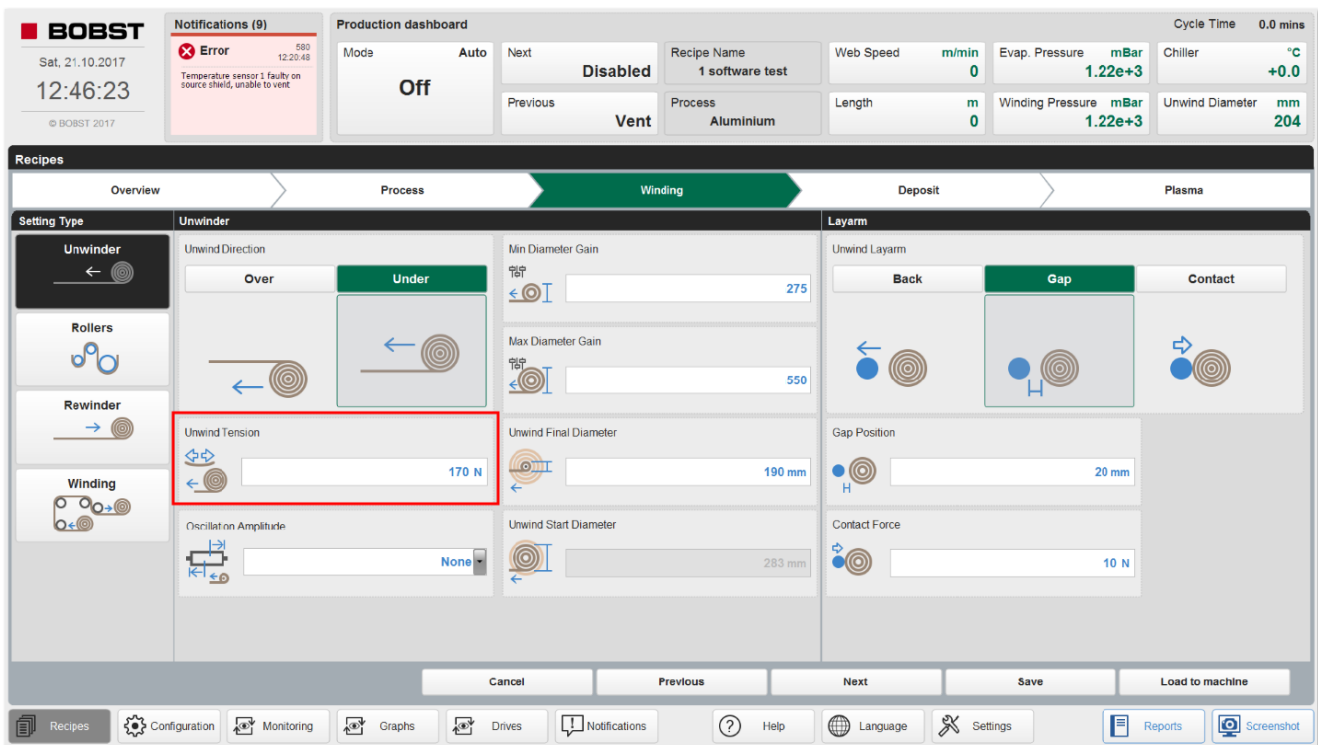
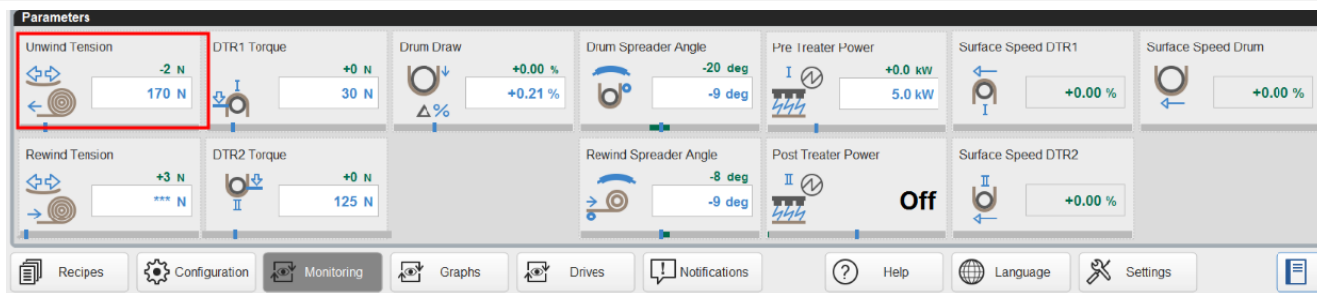


Figure 4.8-1 RECIPES HMI screen

The set tension and actual tension levels will be shown on the MONITORING HMI screen.



The operator can adjust the set-point while the coating process takes place from the HMI MONITORING and DRIVES screens by touching the box section where the set point can be changed.

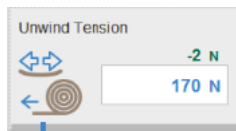


Figure 4.8-2 MONITORING HMI screen – unwind tension section

- The blue characters represent the set point
- The green characters represents the feedback value
- The bar chart at the bottom of the section indicates the actual value shown graphically in a range equal to the full range of the set point.

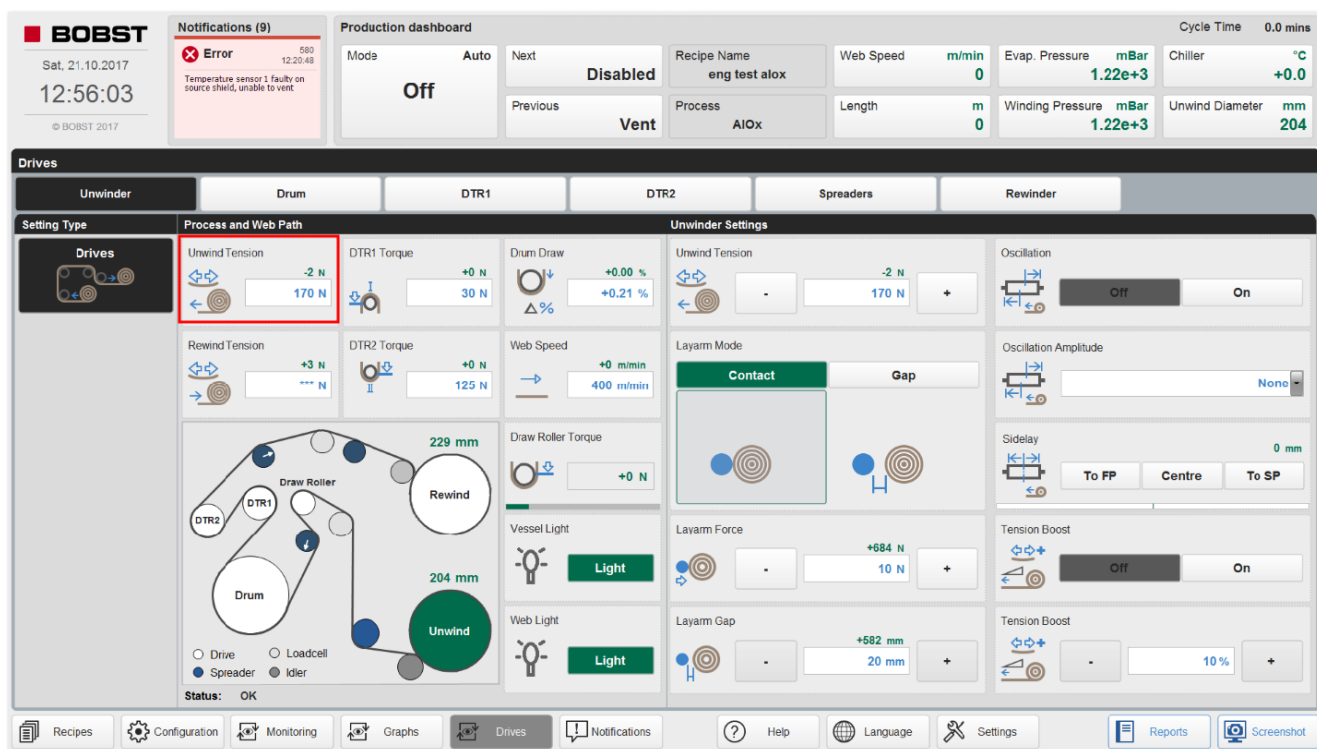


Figure 4.8-3 DRIVES HMI screen – unwind tension section

A pop up numeric box will appear to allow the number to be changed.

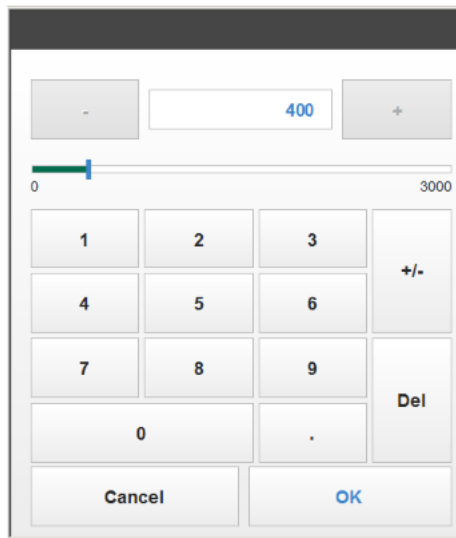


Figure 4.8-4 HMI set point change numeric interface screen

The +/- buttons can be used or the value be changed by the use of the number buttons.

The blue bar indicates the range that the parameter can be set to.

Rewind Tension Level Set Point - Taper or Profile Control

Rewinding at a constant tension may not be desirable for all substrates. For example, when winding thin film, the reel weight coupled with the line tension can tend to crush the reel as it builds up and the weight increases.

Taper tension or profiling are two ways of modifying the tension throughout the diameter build up to suit requirements of the rewind roll.

The change in set point applied could be positive or more usually negative.

The operator has the ability to select profile tension control for this machine in the RECIPE HMI screen.

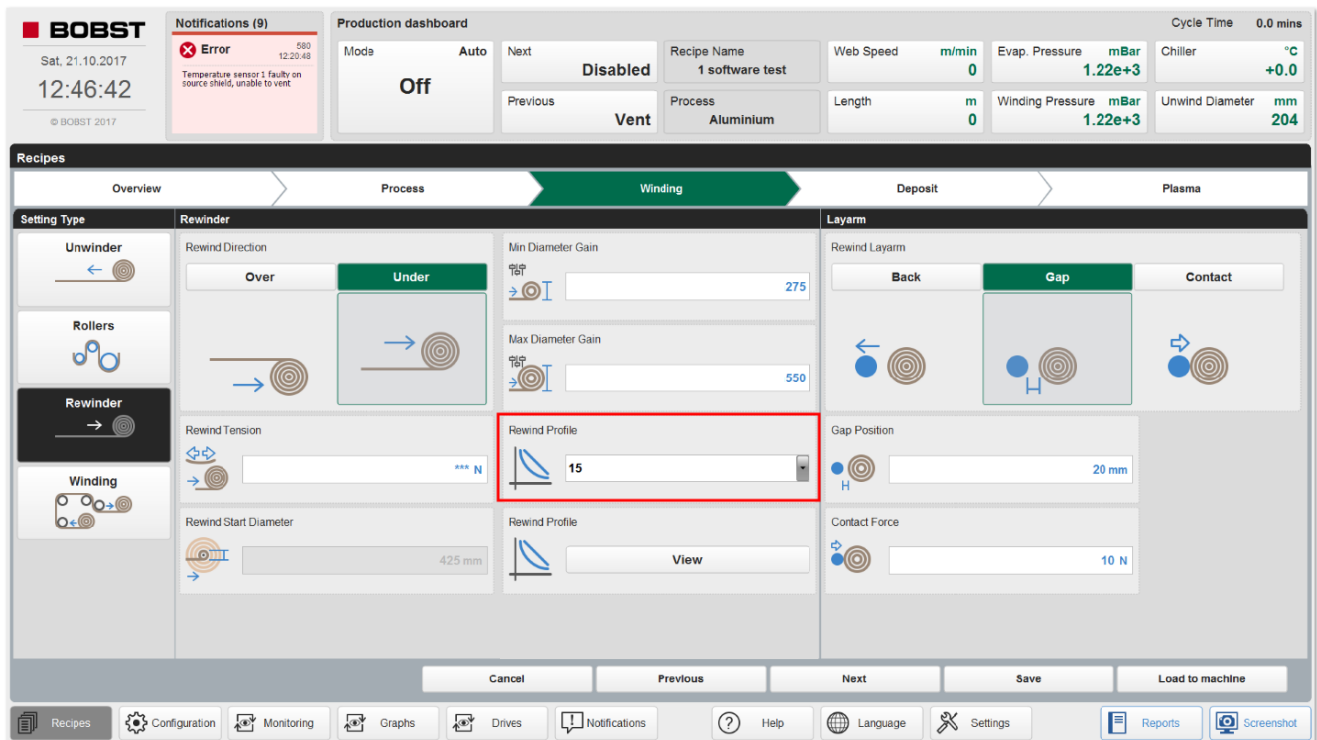


Figure 4.8-5 RECIPES HMI screen with rewind tension set point highlighted.

Touching this section of the screen will give the user the ability to make a selection of the existing available curves.

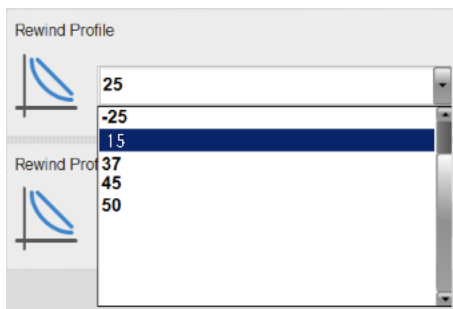


Figure 4.8-6 RECIPE HMI screen – Rewind profile curve selector

The numbers shown are either + or – taper curves.

The curve can be viewed after being selected by touching the VIEW button.

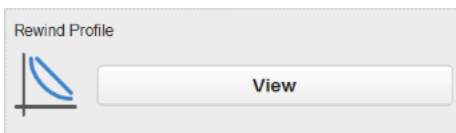


Figure 4.8-7 RECIPE HMI screen – Rewind profile curve VIEW button

Figure 4.8-8

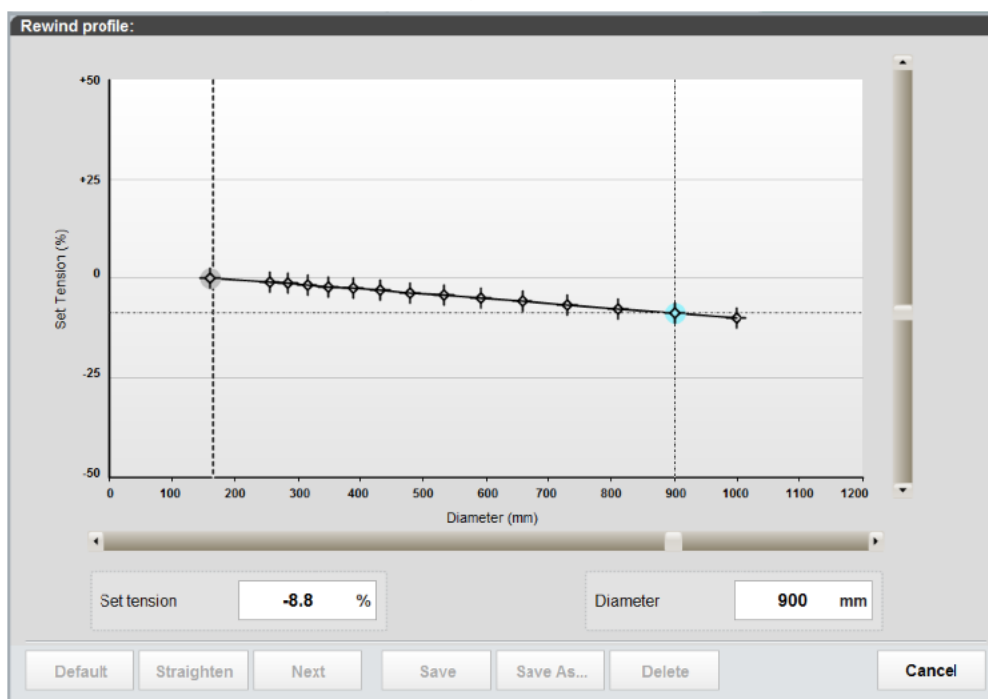


Figure 4.8-9 RECIPE HMI screen – Rewind profile curve after selection from VIEW button.

	Note !
	Modifications to this curve cannot be made at this access level. The curves are edited and saved from the DIAGNOSTICS - MAINTENANCE REWIND PROFILE screen.

When To Use Taper Profile Tension?

For the majority of plastic web materials running in vacuum constant tension control is adequate.

However, with certain materials, such as paper, it has been found that the tension level is required to increase or decrease to give a good rewind reel hardness profile.

When processing paper, it is normal to decrease the tension set point as the roll diameter increases. If it is not used, a corkscrew effect can take place where the roll becomes tighter, the larger that it becomes. The result can be that the roll will telescope sideways.

If there is sideways slipping of the substrate at the rewind roll as it increases in diameter, the set tension should be increased. If the taper is set or profile curve is plotted correctly, the drive system will increase the tension sufficiently to prevent the slippage-taking place. This can be the case in large diameter rolls of PET substrate.

What is Taper?

The term taper is described as a "narrowing of a solid item along its length".

Whilst the design of the winding system is such that set tensions are maintained from, start to finish of a roll.

There are instances, when it is necessary to have a controlled change of tension as the build-up of the substrate on the rewind proceeds.

What is Tension Profile?

The term profile relates to a tension set point value curve that can be created within the computer. The set point will follow this curve as the reel diameter increases.

Creating Rewind Tension Profile Curves



Note !

Higher Login access is required to access the DIAGNOSTICS- MAINTENANCE REWIND PROFILE screen.

To make a new curve or to modify an existing curve, access must be made to the DIAGNOSTICS- MAINTENANCE – REWIND PROFILE function.

A curve file pop-up will be shown to allow the user to select an existing curve.

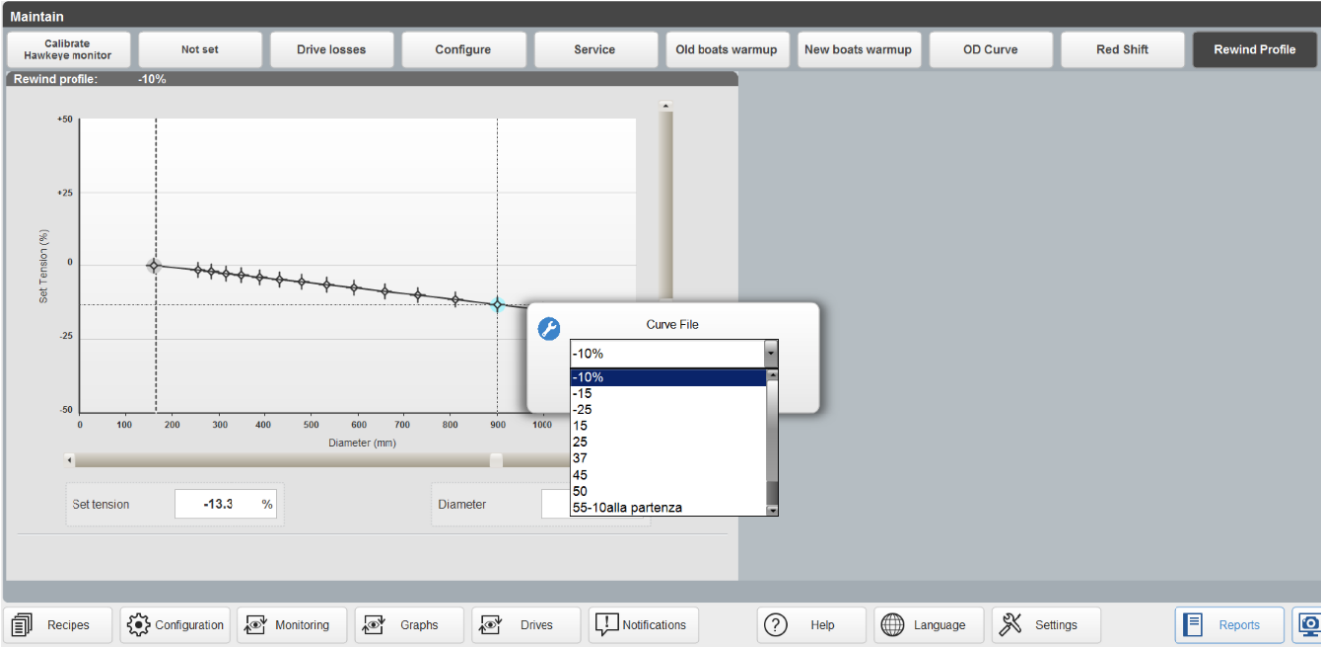


Figure 4.8-10 MAINTAIN- REWIND PROFILE HMI screen – Curve file selection drop down

Select the Default 0% curve from this drop down table.

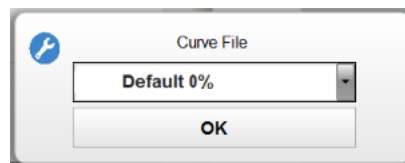


Figure 4.8-11 HMI screen

The default curve will be shown, that can be modified.

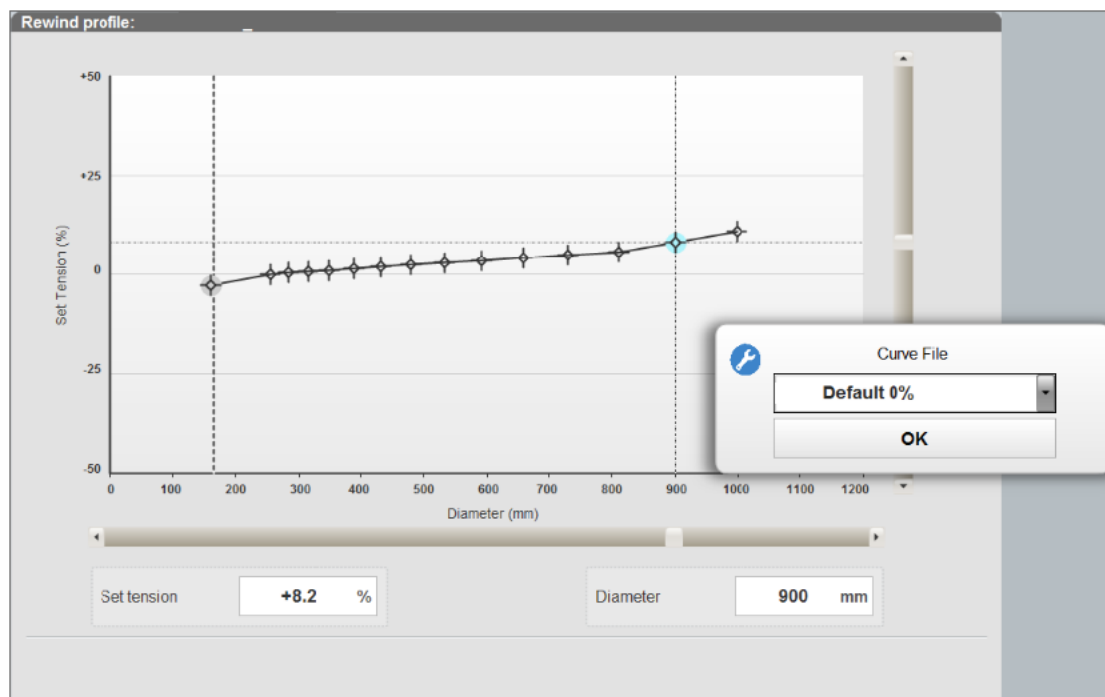


Figure 4.8-12 MAINTAIN- REWIND PROFILE HMI screen – default curve

The curve is plotted by touching the screen on any of the points and modifying either the tension or the diameter value. When these are modified, the point will move to the cross reference point.

Button	Purpose/explanation
Default	Selection of this button will select a Bobst created default curve.
Straighten	Selection of this button can be used to create a curve between two points. For example selecting the first and last points on the curve and pressing this button will create a straight line between them with the other crosshairs spread down its length.
Next	If a point is highlighted, pressing this button will move the blue highlight to the next point. This occurs from left to right.
Save	Selection of this button will save any modifications to the opened curve.
Save As...	Selection of this button will open up a further dialogue box as described later, to allow the curve to be saved under a different name
Delete	Selection of this button will allow the user to delete the current curve
Cancel	Selection of this button closes the screen

Table 4.8-2 Curve editor buttons - Explanation

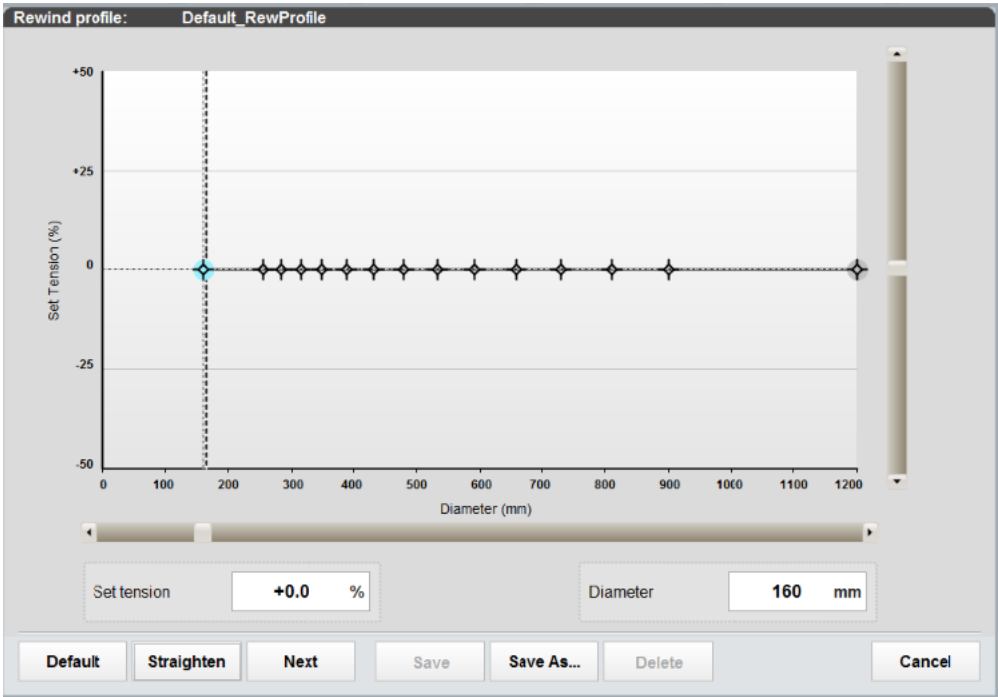


Figure 4.8-13 MAINTAIN- REWIND PROFILE HMI screen – Modified curve

Save As Function

As an example the steps below show a curve called “test” being created.
Select the SAVE As button and the HMI asks the user to name the new curve. In this case we call it “test”

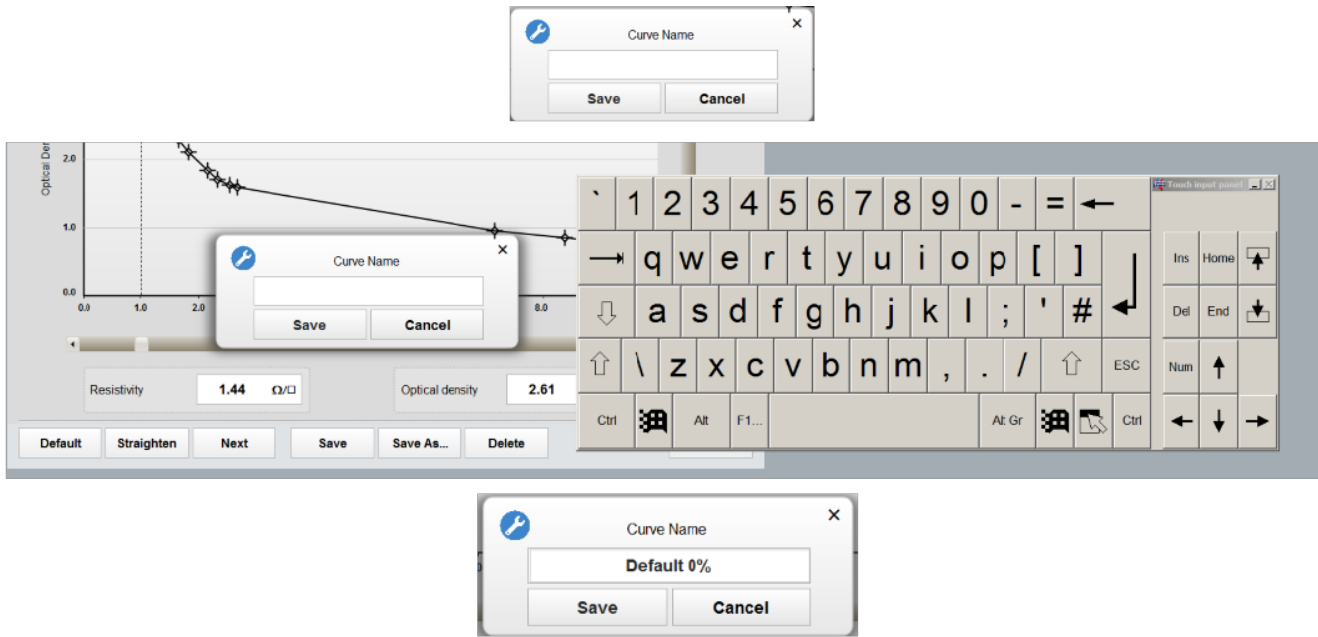


Figure 4.8-14 Example of creating a “saved” OD curve

Delete Function

As an example the steps below show the curve called “test” being deleted.
Select the curve from the dropdown list

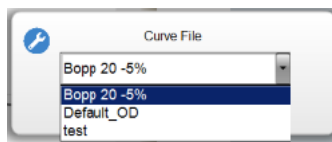


Figure 4.8-15 Exampe of the curve options in the dropdown list

Press the delete button. A pop up screen will be shown asking the user to confirm this action.

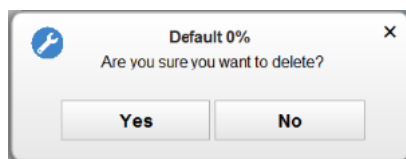


Figure 4.8-16 Example of deleting an OD curve

4.9 Process Roll Set-up Recipes

To assist the user of the equipment, individual recipes can be stored. These are entered using the set up screens.

BOBST
Sat, 21.10.2017
12:40:42
© BOBST 2017

Notifications (9)
Error 580 12:20:48
Temperature sensor 1 faulty on source shield, unable to vent

Production dashboard
Mode: Auto
Next: Disabled
Previous: Vent
Recipe Name: 1 software test-alox
Process: Alubond
Web Speed: 0 m/min
Length: 0 m
Evap. Pressure: 1.22e+3 mBar
Winding Pressure: 1.22e+3 mBar
Chiller: +0.0 °C
Unwind Diameter: 204 mm
Cycle Time: 0.0 mins

Recipes

Recipe Name	Process	Material	Thickness	Width	Deposit	BOBST	Linked
eng test	Aluminium	PET	22 µm	1100 mm	0.13 OD	-	-
1 eng test pet 12	Aluminium	PET	12 µm	1200 mm	3.50 OD	X	-
1 software test	Aluminium	PET	12 µm	1200 mm	2.80 OD	X	-
eng valid	Aluminium	PET	12 µm	1750 mm	4.00 OD	-	-

Favourites All View Save Live Delete Load to machine

Recipes Configuration Monitoring Graphs Drives Notifications Help Language Settings Reports Screenshot

Figure 4.9-1 RECIPES HMI screen – Recipe selection – operator access level

The user can select either “FAVOURITES” or “ALL” existing recipes using the buttons.

BOBST
Sat, 21.10.2017
12:40:42
© BOBST 2017

Notifications (9)
Error 580 12:20:48
Temperature sensor 1 faulty on source shield, unable to vent

Production dashboard
Mode: Auto
Next: Disabled
Previous: Vent
Recipe Name: 1 software test-alox
Process: Alubond
Web Speed: 0 m/min
Length: 0 m
Evap. Pressure: 1.22e+3 mBar
Winding Pressure: 1.22e+3 mBar
Chiller: +0.0 °C
Unwind Diameter: 204 mm
Cycle Time: 0.0 mins

Recipes

Process	No Filter	Material	No Filter	Thickness	No Filter	No Filter	Width	No Filter	Deposit	No Filter	BOBST Recipe	No Filter
Recipe Name												
1 ALUBOND NC		Alubond	PET	12 µm			1200 mm		3.50 OD		X	-
1 software test		Aluminium	PET	12 µm			1200 mm		2.80 OD		X	-
1 software test3		Alubond	PET	12 µm			1200 mm		3.50 OD		X	-
1 software test-alox		AlOx	PET	9 µm			1200 mm		2.80 OD		X	-
12345		Alubond	PET	12 µm			1200 mm		3.50 OD		X	-
150 ALUBOND NC		Alubond	PET	150 µm			770 mm		2.25 OD		-	-
Alux pe		AlOx	LDPE	60 µm			1300 mm		0.13 OD		-	-
alubond bopp		AlOx	LDPE	20 µm			1930 mm		3.00 OD		X	-
alubond po		Alubond	BOPP	12 µm			2430 mm		2.80 OD		X	-
bopp 18u Dark		Alubond	PET	18 µm			1500 mm		2.40 OD		X	-

Favourites All View Save Live Delete Load to machine

Recipes Configuration Monitoring Graphs Drives Notifications Help Language Settings Reports Screenshot

Figure 4.9-2 RECIPES HMI screen – Recipe selection – operator access level

Sort Facility

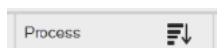


Figure 4.9-3 RECIPES HMI screen - Sort button

This function will sort the files in alphabetical or reverse alphabetical order.

Filter Facility

Filter the list at the top of the screen.

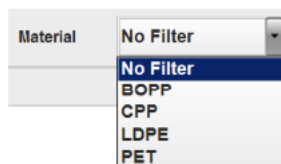


Figure 4.9-4 RECIPES HMI screen - Filter tab

The filter fields are as follows:

- Process: Depending upon the machine specifications and available process, this may be aluminium /Alubond/ Alox
- Material: The type of substrate being processed
- Thickness: Thickness and measurement parameter, microns, mil, gauge, gsm
- Width: Width of the substrate being processed
- Deposit: Product target deposition
- BOBST Recipe: Redefined recipes

Favourites List of Recipes

The Favourites list is defined at supervisor login level to allow operators to quickly select common recipes.

The list shows a maximum of nine recipes, plus the currently selection at the top.

Recipes

Process	No Filter	Material	No Filter	Thickness	No Filter	No Filter	Width	No Filter	Deposit	No Filter	BOBST Recipe	No Filter
Recipe Name												
1 ALUBOND NC		Alubond	PET	12 μm			1200 mm		3.50 OD		X	-
✓ 1 software test		Aluminium	PET	12 μm			1200 mm		2.80 OD		X	-
✓ 1 software test3		Alubond	PET	12 μm			1200 mm		3.50 OD		X	-
✓ 1 software test-alox		AlOx	PET	9 μm			1200 mm		2.80 OD		X	-
✓ 12345		Alubond	PET	12 μm			1200 mm		3.50 OD		X	-
✓ 150 ALUBOND NC		Alubond	PET	150 μm			770 mm		2.25 OD		-	-
✓ Alox pe		AlOx	LDPE	60 μm			1300 mm		0.13 OD		-	-
✓ alubond bopp		AlOx	LDPE	20 μm			1930 mm		3.00 OD		X	-
✓ alubond po		Alubond	BOPP	12 μm			2430 mm		2.80 OD		X	-
✓ bopp 18u Dark		Alubond	PET	18 μm			1500 mm		2.40 OD		X	-

Favourites

All

View

Save Live

Delete

Load to machine

Figure 4.9-5 RECIPES HMI screen - Favourites

ALL List of Recipes

Shows all recipes that have been created.

Editing Recipes

When logged in at the supervisor access level, the VIEW button is replaced by the EDIT button. The SAVE AS and DELETE buttons are also active.

To edit or create a recipe, select one from the list and then press EDIT.

Recipes								
Recipe Name	Process	Material	Thickness	Width	Deposit		BOBST	Linked
✓ eng test alox	AIOx	PET	22 µm	1100 mm	0.13	OD	-	-
✓ 1 eng test pet 12 alubond	Alubond	PET	12 µm	1200 mm	3.50	OD	X	-
✓ 1 software test	Aluminium	PET	12 µm	1200 mm	2.80	OD	X	-
✓ eng valid	Aluminium	PET	12 µm	1750 mm	4.00	OD	-	-

FavouritesAll

EditSave LiveDeleteLoad to machine

RecipesConfigurationMonitoringGraphsDrivesNotificationsHelpLanguageSettingsReportsScreenshot

Figure 4.9-6 RECIPES HMI screen - Change of editing modes

Navigation of Recipes Screens

Use the Previous and Next tabs at the bottom of the screen.

Recipes

OverviewProcessWindingDepositPlasma

Setting Type

Web I

Source II

Gas Wedge III

General

Web Type

PET

Web Density

1.38 g/cm3

Thickness

22 µm

Width

Web Width

1100 mm

Adaptive Width

OffOn

Adaptive Width Database

Adaptive Width Disabled

CancelPreviousNextSaveLoad to machine

Figure 4.9-7 RECIPES HMI screen - Navigation

Touching the NEXT button will cycle through the screens in the following order:

- PROCESS – WEB I
- SOURCE II
- GAS WEDGE III
- WINDING - UNWINDER.

Entering Data

This bar indicates minimum and maximum range of the parameter data entry.

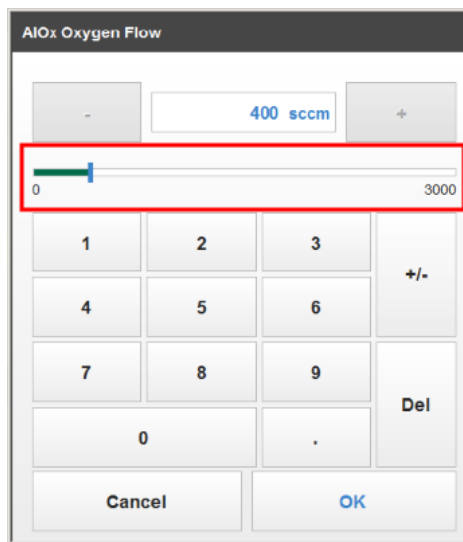


Figure 4.9-8 RECIPES HMI screen - Pop up Numeric keyboard



Figure 4.9-9 RECIPES HMI screen - Pop up keyboard

Overview Screen

The overview tab allows quick access to the common parameters that may change from roll to roll.

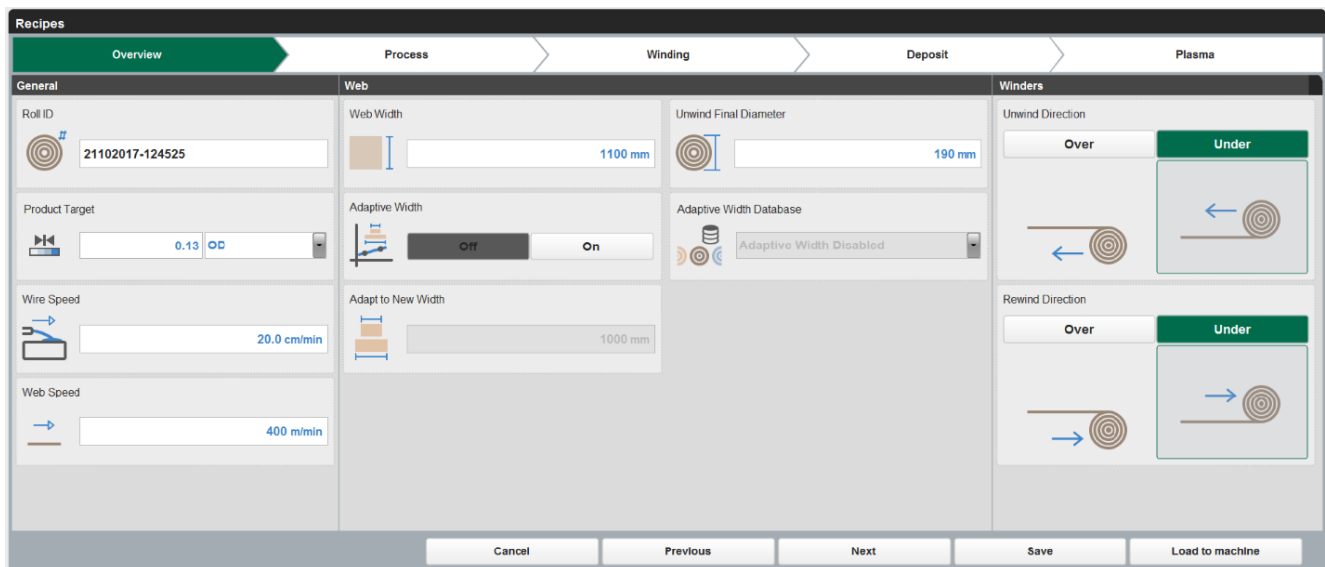


Figure 4.9-10 RECIPES HMI screen - OVERVIEW

Parameter	Description
Roll ID	Used to find an archived report at a later date.
Product Target	Sets the deposition target and measuring parameter

Table 4.9-1 OVERVIEW screen explanations

Parameter	Description
Wire Speed (cm/min)	Required aluminium wire feeder speed
Web Speed (m/min)	Sets required Line Speed
Web Width (mm)	Important information for the machine to select the correct number of evaporators and aluminium wire feeders.
Adaptive Width	Selects the adaptive width function usage
Adapt to New Width	Adjusts the drive set point to suit the new width.
Unwind Final Diameter	Desired diameter that the user wants to stop the winder at to leave the minimum amount of waste material. Important information for the drive system
Adaptive Width database	Loads the set of tension setting data from the BOBST created machine database file.
Unwind Direction	Sets shaft winding direction
Rewind Direction	Sets shaft winding direction

Table 4.9-1 OVERVIEW screen explanations

Process

The process tab shows the process settings for the recipe. The process settings are split into types. Select a type to show its parameters:

- Web
- Source
- Gas Wedge

Process: Web Settings

The screenshot displays the 'RECIPES HMI screen - PROCESS - WEB' interface. The top navigation bar includes tabs for Overview, Process (highlighted in green), Winding, Deposit, and Plasma. Below this, a 'Setting Type' sidebar on the left lists 'Web' (selected), 'Source', and 'Gas Wedge'. The main content area is divided into two columns: 'General' and 'Width'.
 In the 'General' column:
 - 'Web Type' is set to 'PET'.
 - 'Web Density' is set to '1.38 g/cm3'.
 - 'Thickness' is set to '22 µm'.
 In the 'Width' column:
 - 'Web Width' is set to '1100 mm'.
 - 'Adaptive Width' is a toggle switch currently set to 'Off'.
 - 'Adaptive Width Database' is set to 'Adaptive Width Disabled'.
 At the bottom of the screen, a row of buttons includes 'Cancel', 'Previous', 'Next', 'Save', and 'Load to machine'.

Figure 4.9-11 RECIPES HMI screen - PROCESS - WEB

Parameter	Description
Web Type	Type of substrate that can be selected from the drop down list.
Web Density (g/cm3)	Important information for the drive system
Thickness	Important that the drive system has this information to allow automatic diameter stopping.
Web Width (mm)	Important information for the machine to select the correct number of evaporators and aluminium wire feeders.
Adaptive Width ON/OFF	Selects if the function is switched on or off.
Adaptive Database	Loads the set of tension setting data from the BOBST created machine database file.
Adapt to New Width	Adjusts the drive set point to suit the new width.

Table 4.9-2 Process web screen explanations

Process: Source Settings

The screenshot displays the 'PROCESS - SOURCE' settings screen. The interface includes a top navigation bar with tabs for Overview, Process (selected), Winding, Deposit, and Plasma. Below this, there's a 'Setting Type' sidebar with options for Web, Source (selected), and Gas Wedge. The main area is divided into sections: General (Process Type: AIOx, AIOx Gas Flow: 400 sccm, Alternate Boats: On), Wire Speed (20.0 cm/min), Shutter Open % Web Speed (50%), Shutter Close % Web Speed (47%), and Chiller (Drum Temp: -5 °C, DTR1 Cooling: Off/On). At the bottom, there are buttons for Cancel, Previous, Next, Save, and Load to machine.

Figure 4.9-12 RECIPES HMI screen - PROCESS - SOURCE

Parameter	Description
Process Type	Allows selection of one of the process modes delivered with the machine.
Alox Gas Flow	Sets the base level of gas flow
Alternate Boats	Switches the mode of control between all or alternative boats
Wire Speed	Required aluminium wire feeder speed
Shutter Open % Web Speed	Percentage of set line speed that the shutter opens

Table 4.9-3

Process source screen explanations

Parameter	Description
Shutter Close % Web Speed	Percentage of set line speed that the shutter closes.
Drum Temp (C)	Sets chiller unit feeding process drum & roller temperature
DTR1 Cooling On/Off	Selecting this feature allows cooling and heating of the DTR1 (WCR) roller to take place. If not selected, the rewind roll maybe hotter

Table 4.9-3

Process source screen explanations

Process: Gas Wedge Settings

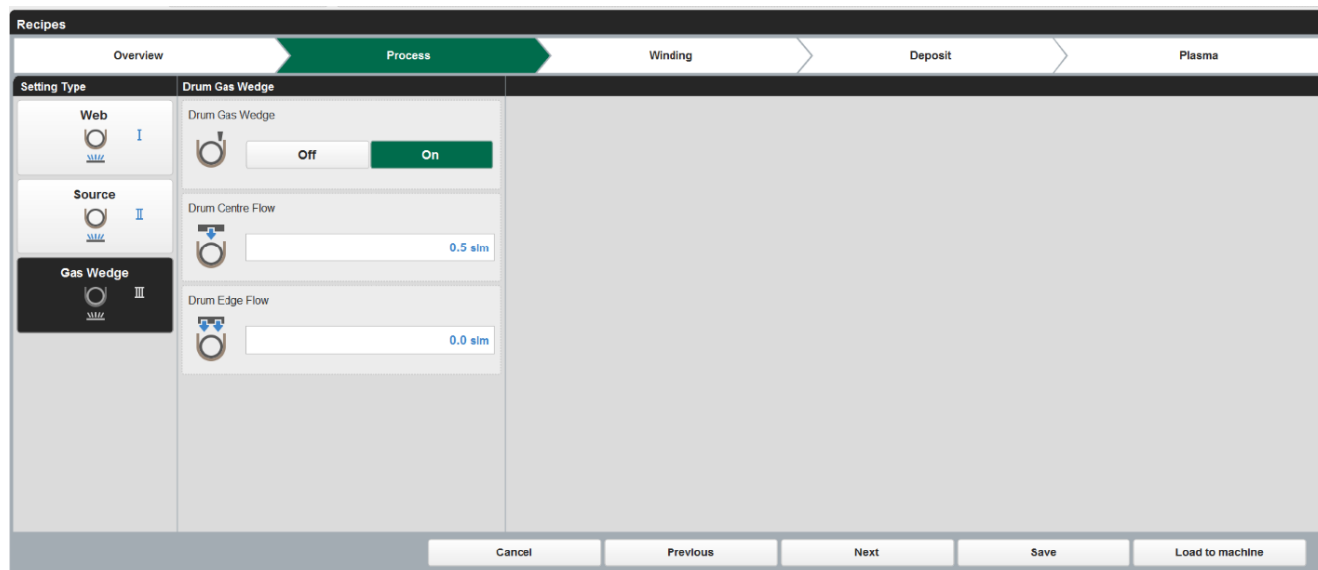


Figure 4.9-13 RECIPES HMI screen - PROCESS - GAS WEDGE

Parameter	Description
Drum Gas Wedge	Selection of use – On/Off
Drum Centre Flow (slm)	Allows the air flow set point to be set for this area of the process drum gas wedge system
Drum Edge Flow (slm)	Allows the air flow set point to be set for this area of the process drum gas wedge system

Table 4.9-4 Process gas wedge screen explanations

Winding

The winding tab shows the winding settings for the recipe. The winding settings are split into types. Select a setting type to show its parameters:

- Unwinder
- Rollers
- Rewinder
- Winding

Winding: Unwinder Settings

The screenshot shows the 'Winding' tab selected in the 'Recipes' HMI. The 'Unwinder' section is active, displaying the following settings:

- Setting Type:** Unwinder (selected), Rollers, Rewinder, Winding.
- Unwind Direction:** Over (deselected), Under (selected).
- Min Diameter Gain:** 275.
- Max Diameter Gain:** 550.
- Unwind Tension:** 170 N.
- Oscillation Amplitude:** None.
- Unwind Final Diameter:** 190 mm.
- Unwind Start Diameter:** 283 mm.
- Layarm:** Back (deselected), Gap (selected), Contact (deselected).
- Gap Position:** 20 mm.
- Contact Force:** 10 N.

Navigation buttons at the bottom include: Cancel, Previous, Next, Save, and Load to machine.

Figure 4.9-14 RECIPES HMI screen - WINDING - UNWINDER

Parameter	Description
Unwind Direction	Sets roll winding direction
Unwind Tension (newtons)	Sets Tension set point
Oscillation Amplitude	Allows the user to select the amount of movement from the drop down selection tab
Min Diameter Gain	Sets the drive response to changes in tension at the roll minimum diameter
Max Diameter Gain	Sets the drive response to changes in tension at the roll maximum diameter
Unwind Final Diameter (mm)	This is the desired diameter that the user wants to stop the winder at to leave the minimum amount of waste material. Important information for the drive system
Unwind Start Diameter (mm)	Important information for the drive system. Only required if the layarm cannot measure the diameter (for example, when side cheeks are installed).
Unwind Layarm	Selects the mode of operation in either gap or contact
Gap Position (mm)	Allows the selection of the gap between the core and the layarm final roller.
Contact Force (newtons)	Allows the amount of contact off the layarm final roller onto the core

Table 4.9-5 Winding - unwinder screen explanations

Winding: Rollers Settings

Figure 4.9-15 RECIPES HMI screen - WINDING - ROLLERS

Parameter	Description
Drum Control mode	Selects the mode of operation - draw/tension
Drum Torque (newtons)	Set point value entered here
Drum Draw (%)	Set point value entered here
DTR1 Control mode	Selects the mode of operation - draw/tension
DTR1 Torque (newtons)	Set point value entered here
DTR1 Draw (%)	Set point value entered here
DTR2 Control mode	Selects the mode of operation - draw/tension
DTR2 Torque (newtons)	Set point value entered here
DTR2 Draw (%)	Set point value entered here

Table 4.9-6 Winding - rollers screen explanations

Winding: Rewinder Settings

Figure 4.9-16 RECIPES HMI screen - WINDING - REWINDER

Parameter	Description
Rewind Direction	Sets roll winding direction
Rewind Tension (newtons)	Sets Tension set point
Rewind Start Diameter (mm)	Important information for the drive system
Min Diameter Gain	Sets the drive response to changes in tension at the roll minimum diameter
Max Diameter Gain	Sets the drive response to changes in tension at the roll maximum diameter
Rewind Profile – Curve select	Allows the user to select one of the existing curves
Rewind Profile view	Allows the user to view the curve to ensure that the correct curve has been selected
Oscillation Amplitude	Allows the user to select the amount of movement from the drop down selection tab
Rewind Layarm Mode	Selects the mode of operation in either gap or contact
Gap Position (mm)	Allows the selection of the gap between the core and the layarm final roller.
Contact Force (newtons)	Allows the amount of contact off the layarm final roller onto the core

Table 4.9-7 Winding - rewinder screen explanations

Winding: Winding Settings

The screenshot displays the 'Winding' HMI screen with a sidebar on the left containing icons for Unwinder, Rollers, Rewinder, and Winding (selected). The main area is divided into four panels:

- General:** Includes 'Web Speed' (400 m/min) and 'DTR1 Non Contact' mode (set to 'Off').
- Spreaders:** Includes 'Drum Spreader' and 'Rewind Spreader', both set to '-9 deg'.
- Web Ramp Time:** Includes 'Acceleration' and 'Deceleration', both set to '60 s'.

Figure 4.9-17 RECIPES HMI screen - WINDING

Parameter	Description
Web Speed (m/min)	Sets required Line Speed
DTR1 Non Contact mode	Allows the user to select the web path to be used, which will determine if the substrate bypasses the DTR1 roller
Drum Spreader (deg)	Sets the apex angle of the spreader roller in this location
Rewind Spreader (deg)	Sets the apex angle of the spreader roller in this location

Table 4.9-8 Winding screen explanations